
Interpreting ResNet-based CLIP via Neuron-Attention Decomposition

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 We present a novel technique for interpreting the neurons in CLIP-ResNet by
2 decomposing their contribution into computation paths. This approach, analyzing
3 all pairwise combinations of the neurons and attention heads of CLIP-ResNet’s
4 attention pooling layer, yields more useful information than previous interpretability
5 methods. Specifically, unlike individual neurons, our neuron-head pairs can
6 be approximated by a single direction in CLIP-ResNet’s image-text embedding
7 space. Leveraging this insight, we associate text with each neuron-head pair and
8 create refined training-free semantic segmentation maps, outperforming previous
9 methods for CLIP-ResNet. Additionally, our analysis shows that a sparse set of the
10 neuron-head pairs contribute extremely significantly to the output value, and that
11 neuron-head pairs, while polysemantic, sometimes represent sub-concepts of their
12 corresponding neurons. Finally, we apply our knowledge of neuron-head pairs to
13 two dataset distribution shift case studies. Our results demonstrate the viability of
14 neuron-head pairs as an improved unit of measurement for interpretability, and that
15 such improvements meaningfully increase performance in downstream tasks.

16 1 Introduction

17 Interpreting the hidden components in pre-trained deep neural networks, by tracing their contribution
18 to the model output, allows us to detect model limitations and useful sub-computations that can be
19 repurposed for multiple downstream tasks (Lindsey et al. [2025], Sharkey et al. [2025]). Recently,
20 such approaches were applied to CLIP - a widely used class of image encoders (Gandelsman et al.
21 [2024b], Bhalla et al. [2024], Gandelsman et al. [2024a]). These approaches unlocked various capabilities - discovery of spurious correlations in the model output, automated generation of adversarial attacks, and even reuse of the interpreted model components for segmentation.

24 While CLIP interpretability work showed promising results for the vision transformer-based variants
25 (CLIP-ViT), such existing methods do not readily extend to the ResNet counterparts (CLIP-ResNet).
26 Most of these methods rely on a decomposition of the output into a sum of per-layer contributions,
27 which is not possible for CLIP-ResNet, as, despite its additive residual connections, each layer is
28 followed by a non-linearity. Moreover, CLIP-ViT interpretability methods rely on the existence of
29 attention blocks and a special class token, but CLIP-ResNet models have convolutions and final
30 attention pooling instead. These architectural differences make existing methods not applicable.

31 To address this gap, we introduce a new approach that provides a fine-grained decomposition of the
32 ResNet-based CLIP model outputs into individual contributions of neurons in the last layers and the
33 following attention-pooling heads. More specifically, we show that the output of CLIP-ResNet is a
34 sum over computation paths, going from individual neurons in the last layer, in parallel through all the

⁰Finalized version will likely be on arXiv!

attention heads in the attention pooling layer, to the output. As each such neuron-head contribution lives in the joint image-text space, it can be compared and interpreted via text.

We analyze the contributions of each neuron-head pair to the output and find that, unlike CLIP-ResNet’s individual neurons, neuron-head pairs can be approximated by a *single direction* in the joint image-text embedding space. Additionally, we find that a fixed sparse set of neuron-head pairs comprises most of the output value (mean-ablating a fixed bottom 80% of neuron-head pairs decreases ImageNet classification accuracy by only 5%). We run analogous experiments for neuron contributions and find, and, furthermore, we discover that some neuron-head pairs for a given neuron encompass sub-concepts of the concepts that the neuron represents (e.g., a ‘butterfly’ neuron can be decomposed into a ‘butterfly clothing’ neuron-head pair and other sub-concepts).

Leveraging our approximation of each neuron-head pair with a single direction, we can rank each pair by its similarity to text representations. We apply this to associate each class with a set of k neuron-head pairs, and use this association for dense semantic segmentation. In more detail, for each class, we obtain segmentation logits by aggregating spatial maps corresponding to the class’s respective k neuron-head pairs. Notably, we outperform previous methods for training-free semantic segmentation using CLIP-ResNet.

Additionally, we explore use cases of neuron-head pairs in discovering and monitoring dataset distribution shifts. We showcase how sub-concept neuron-head pairs can be used for identifying fine-grained image set differences, and how neuron-attention contributions can be used to track distribution shifts across a dataset.

In summary, we propose neuron-attention decomposition as an improved interpretability method to automatically label the components of CLIP-ResNet’s vision encoder with text. We find evidence that our decomposition is favorable in comparison to decomposing solely neurons or solely attention heads, and apply this decomposition to several relevant case studies. Broadly, this shows the viability of neuron-head pairs as an improved unit of measurement of which to study the internal mechanisms of models and of which to improve model capabilities.

2 Related Work

Contrastive vision-language models. Models like CLIP (Radford et al. [2021]) and its variants (Jia et al. [2021], Zhai et al. [2023]) are trained on massive web-based datasets of images and their captions to learn meaningful image representations. This pre-training enables zero-shot capabilities for downstream tasks like OCR, geolocalization, and classification (Wortsman [2023]). These models are also used as backbones for other systems such as LLaVA (Liu et al. [2023]), 3-D learning (Zhu et al. [2023]), and image generation (Ramesh et al. [2021], Rombach et al. [2022]).

Interpreting vision models. Mechanistic interpretability is a field of research that seeks to understand the inner workings of neural networks by analyzing fundamental model components and computation paths. Early mechanistic interpretability discoveries for vision models include the attribution of high-level concepts to intermediate model neurons (Bau et al. [2017]), and curve detectors and circuits (Cammarata et al. [2020]). Similar to us, a body of more recent work aims to automatically label vision model components with text (Hernandez et al. [2022], Bills et al. [2023], Oikarinen and Weng [2024]).

Interpreting CLIP. Several recent works investigate CLIP’s embedding space with techniques like sparse coding (Bhalla et al. [2024]) and factor rotation (Zhao et al. [2025]), aiming to identify human-interpretable concepts. Most closely to us, previous work analyzing CLIP-ViT’s output decomposition (Gandelsman et al. [2024a], Gandelsman et al. [2024b]) made use of the additive linearity of a ViT’s residual stream to look at the contribution of individual layers and their components to the embedding space, and relied on the joint output image-text space to interpret such components with text. Differently from these methods, we focus on ResNet-based models for which the contributions of the early layers are not additive to the output (due to a ReLU nonlinearity follows each residual connection). However, we *are* able to interpret CLIP-ResNet’s last convolutional block and its attention pooling, where we propose a more fine-grained view of its inner workings by analyzing neuron-head pairs instead of solely individual neurons or solely individual attention heads.

86 3 Methodology

87 We start by presenting CLIP-ResNet’s architecture and deriving the decomposition of its image
88 representation. We use this decomposition to interpret the contributions of individual pairs of neurons
89 and attention heads in later sections.

90 3.1 CLIP-ResNet preliminaries

91 **Contrastive pre-training.** CLIP is trained via a contrastive loss that aligns the representations of its
92 image encoder M_{image} with the representations of its text encoder M_{text} in a shared image-text latent
93 space $\mathbb{R}^d$. Specifically, over massive web-based datasets, the two encoders are trained together to
94 maximize the cosine similarity for matching image-text pairs (I, t) :

$$\text{sim}(I, t) = \frac{\langle M_{\text{image}}(I), M_{\text{text}}(t) \rangle}{\|M_{\text{image}}(I)\|_2 \|M_{\text{text}}(t)\|_2}. \quad (1)$$

95 **Zero-shot classification.** To perform image classification, each class name c_j is mapped to some
96 template (e.g. “A photo of a {class}”) and encoded by the text encoder as $M_{\text{text}}(\text{template}(c_j))$ (for
97 simplicity, we will omit the template notation). The classification prediction for an image I is the
98 class c_i whose text representation $M_{\text{text}}(c_j)$ is most similar to the image representation $M_{\text{image}}(I)$.

99 **CLIP-ResNet.** The CLIP-ResNet image encoder is a traditional ResNet network [He et al., 2016],
100 composed of sequential *residual blocks* with an average pooling replaced by attention pooling
101 (Radford et al. [2021]). While CLIP is often trained with ViT as the image encoder backbone,
102 CLIP-ResNet is competitive in performance across various benchmarks. However, the internal
103 mechanisms of CLIP-ResNet are underexplored in comparison to CLIP-ViT, and existing methods
104 are not applicable due to architectural differences. We explain this difference and our proposed
105 method to bridge this gap in the next section.

106 **CLIP-ResNet architecture.** CLIP-ResNet’s residual stream is not linear, as the additive residual
107 connection within each residual block is followed by a nonlinear ReLU activation. Thus, we focus
108 only on the final residual block’s input to the attention pooling layer, which we *can* decompose
109 linearly. Formally, given an input image I , let $Z(I)$ be the output of the last convolutional layer in
110 the model, and let $Z'(I)$ be $Z(I)$ after prepending a class token and adding positional embedding.
111 CLIP-ResNet’s image representation is the $Z'(I)$ with an attention pooling applied to it:

$$M_{\text{image}}(I) = \text{AttnPool}(Z'(I)). \quad (2)$$

112 More specifically, the dimensionality of $Z(I)$ is $C \times H' \times W'$ where C is the number of *neurons*
113 (post-ReLU per-location activations in the final convolutional block) and H' and W' are the spatial
114 feature map dimensions. To form $Z'(I) \in \mathbb{R}^{(K+1) \times C}$, $Z(I)$ is first flattened into $K = H'W'$ image
115 tokens $\{z_i\}_{i \in \{1, \dots, K\}}$, $z_i \in \mathbb{R}^C$, then a *class token* $z_0 = \frac{1}{K} \sum_{i=1}^K z_i$ is prepended to this sequence,
116 and finally, a learned positional embedding is added to all $K + 1$ tokens. Attention pooling is
117 implemented as a standard transformer multi-head attention module, with the exception being that
118 the class token is used as the sole output – only the first row of the attention matrix is projected to
119 form the image representation $M_{\text{image}}(I)$.

120 3.2 Decomposition into neurons, heads, and tokens

121 **Decomposition into attention heads and tokens.** Following Elhage et al. [2021] and leveraging the
122 fact that only the class token is returned by attention pooling, we can write the image representation
123 as a sum over H attention heads of the attention-pooling layer and $K + 1$ tokens:

$$M_{\text{image}}(I) = \text{AttnPool}([z_0, \dots, z_K]) = \sum_{h=1}^H \sum_{i=0}^K a_i^h(I) z_i W_{VO}^h \quad (3)$$

124 where $W_{VO}^h \in \mathbb{R}^{C \times d}$ are transition matrices (the OV matrices) and a_i^h is a weight that denotes how
125 much the class token attends to the i -th token ($\sum_{i=0}^K a_i^h = 1$).

PC(s)	Accuracy (%)
(Baseline)	70.7
$\{\hat{r}^{n,h}\}$	70.7
$\{\hat{r}_1^n\}$	66.9
$\{\hat{r}_1^n, \hat{r}_2^n\}$	69.0
$\{\hat{r}_1^n, \hat{r}_2^n, \hat{r}_3^n\}$	69.7
$\{\hat{r}_1^n, \hat{r}_2^n, \hat{r}_3^n, \hat{r}_4^n\}$	70.0

Table 1: **Accuracy for reconstruction from principal components (RN50x16).** Unlike neuron representations, neuron-attention representations match the baseline accuracy using only one direction for reconstruction.

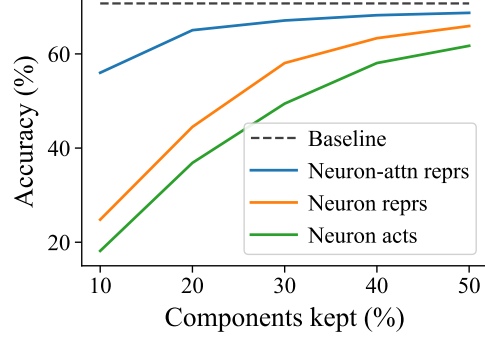


Figure 1: **Mean-ablation accuracy (RN50x16).** Mean-ablating all but a fixed set of components shows there is a sparse set of neuron-head pairs that is very significant.

126 **Decomposition into neuron-head pairs.** Each row of the W_{VO}^h matrix corresponds to one neuron.
127 That means, for any given head and token, we can rewrite:

$$z_i W_{VO}^h = \sum_{n=1}^C z_i W_{VO}^{n,h} \quad (4)$$

128 where the superscript n denotes the n -th row of W_{VO}^h . Substituting into (3) and swapping the sum
129 yields:

$$M_{\text{image}}(I) = \sum_{n=1}^C \sum_{h=1}^H \sum_{i=0}^K r_i^{n,h}(I), \quad r_i^{n,h} = a_i^h(I) z_i W_{VO}^{n,h}. \quad (5)$$

130 $r_i^{n,h}$ denotes the d -dimensional contribution to the output from the class token’s attention to token i
131 through head h , projected by the row of W_{VO}^h that corresponds to neuron n .

132 The image representation lives in the joint image-text embedding space. Therefore, each neuron-
133 attention contribution $r_i^{n,h}$ obtained by summing over tokens, and each neuron contribution r^n ,
134 obtained by summing over tokens *and* heads, lives in the same image-text space, which allows us to
135 compare them to text. We will use this property for automatically labeling neurons and neuron-head
136 pairs in later sections. Summing over only neurons, instead, gives the decomposition r_i^h , which
137 corresponds to the spatial patches at locations $i \in \{1, HW\}$. This allows us to compute the similarity
138 to text for each image location represented in the decomposed class token, and we use this property
139 to form our per-token segmentation map in Section 5.2.

140 **Comparison to existing decompositions.** Previous work on CLIP-ViT has studied the direct effect
141 of decomposition across attention heads and tokens [Gandelsman et al., 2024b], analogous to our
142 decompositions r_i^h and r^h , as well as the second-order effect of decomposition into neurons [Gan-
143 delsman et al., 2024a], analogous to our decomposition r^n . We show that our approach overcomes
144 two main limitations of these methods, when applied to CLIP-ResNet: 1) there are a small amount
145 of attention heads relative to possible concepts that CLIP learns, meaning that each head encodes
146 multiple concepts, ultimately making them less interpretable [Lecomte et al., 2024], and 2) the
147 second-order effect of neurons disregards the independent nature of attention heads in the multi-head
148 attention layer, meaning r^n encompasses all paths through the network – which, as we will show
149 next, captures multi-dimensional conceptual structure that is yet again difficult to interpret.

150 4 Analysis

151 We quantitatively evaluate contributions of individual components in our neuron-attention decompo-
152 sition, and show their benefits over other decompositions – these contributions are approximately
153 rank-1 and sparse. Additionally, we qualitatively investigate the top-activating images for selected
154 components and discover that some neuron-head pairs for a given neuron represent sub-concepts of
155 that neuron. We later use these observations to produce semantic segmentation (Section 5.1) and
156 monitoring distribution shift between image sets (Section 5.2).

PC(s)	Accuracy (%)
(Baseline)	70.7
$\{\hat{r}^{n,h}\}$	70.7
$\{\hat{r}_1^n\}$	65.9
$\{\hat{r}_1^n, \hat{r}_2^n\}$	68.3
$\{\hat{r}_1^n, \hat{r}_2^n, \hat{r}_3^n\}$	69.1
$\{\hat{r}_1^n, \hat{r}_2^n, \hat{r}_3^n, \hat{r}_4^n\}$	69.6

Table 2: **Another figure? (polysematicity?) (or a small figure earlier to visualize the neuron-attention decomposition?).**

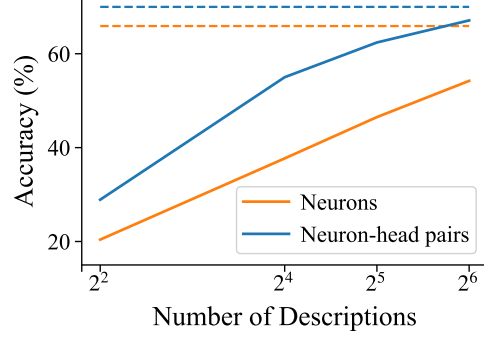


Figure 2: **Reconstruction from sparse text representations (RN50x16).** We compare sparse text decompositions for neurons and neuron-head pairs while varying description set sizes.

157 4.1 Quantitative analysis

158 We study the properties of $r^{n,h}(I)$ and $r^n(I)$. Notably, we find that individual $r^{n,h}(I)$ can be
159 approximated by a single direction in the image-text space, while approximating $r^n(I)$ with only
160 one linear direction causes a significant drop in reconstruction fidelity. Moreover, we find that
161 (neuron, head) pairs are contributing more sparsely than neurons-only – mean-ablating all but a
162 subset of neurons and neuron-head pairs shows a sparser set of top-contributing neuron-head pairs in
163 comparison to neurons.

164 **Experimental setting.** We measure the performance of zero-shot classification on the ImageNet
165 (Deng et al. [2009]) validation dataset after various ablations to quantify the resulting change in
166 the representation. We collect $r^{n,h}(I)$ and $r^n(I)$ representations over the set $\mathcal{D}$, which comprises
167 1000 images from the ImageNet test dataset. We use our collected representations to compute
168 singular value decomposition, from which we obtain our principal components, and to compute mean
169 contributions, which we use for our mean-ablation (Nanda et al. [2023]) experiments. Additionally,
170 after obtaining each neuron-head pair’s principal component, we further decompose it using a sparse
171 coding technique and evaluate its sparsity-performance tradeoff.

172 **Neuron-attention contributions are one-dimensional.** $r^{n,h}(I)$ can be approximated by a single
173 direction in the joint embedding space $\hat{r}^{n,h}$. We approximate $r^{n,h}(I)$ for any given image I with
174 $x^{n,h}(I) \hat{r}^{n,h} + b^{n,h}$, where $x^{n,h}(I)$ is the coefficient obtained from the projection norm of $r^{n,h}(I)$
175 onto $\hat{r}^{n,h}$, and $b^{n,h}$ is the mean of all $r^{n,h}(I)$ contributions over all images $I \in \mathcal{D}$. As shown in
176 Table 1, replacing $r^{n,h}(I)$ with this approximation results in no decrease in ImageNet classification
177 accuracy.

178 **Neuron-attention contributions are sparse.** Keeping only 20% of neuron-head pair contributions
179 over ImageNet (while mean-ablating the rest) results in only a $\sim 5\%$ decrease in classification
180 accuracy. We sort the $C \times H$ neuron-head pairs by the mean of their top percentile norms over $\mathcal{D}$,
181 and observe the same high-scoring pairs $\mathcal{P}^*$ tend to be consistently important across the dataset.
182 Decomposing $M_{\text{image}}(I)$ into individual $r^{n,h}$ contributions, we keep only the $r^{n,h}$ whose neuron-head
183 pairs are in $\mathcal{P}^*$ and mean-ablate the rest. We construct $\mathcal{P}^*$ with varying top percentage norms (at
184 10% increments) and measure the resulting classification accuracy in Figure 1. As shown, there is a
185 sparse set of neuron-head pairs that contribute most of the output value.

186 **Sparse text-based decomposition of neuron-attention directions.** Following Gandelsman et al.
187 [2024a], we experiment with using orthogonal matching pursuit (Pati et al. [1993]) to further
188 decompose each $\hat{r}^{n,h}$ direction into sparse text components. Formally, we use a sparse set of text
189 components $\{t_j\}_{j \in \{1,m\}}$ to approximate $\hat{r}^{n,h} \approx \sum_{j=1}^m \gamma_j^{n,h} M_{\text{text}}(t_j)$, where $\gamma_j^{n,h}$ is a non-zero
190 scalar coefficient. In our experiments, the initial pool of text descriptions is composed of the 30,000
191 most common English words. We vary the sparse set size m and present classification accuracy
192 performance on ImageNet, after replacing $\hat{r}^{n,h}$ with our sparse approximation in Figure 2. As shown,

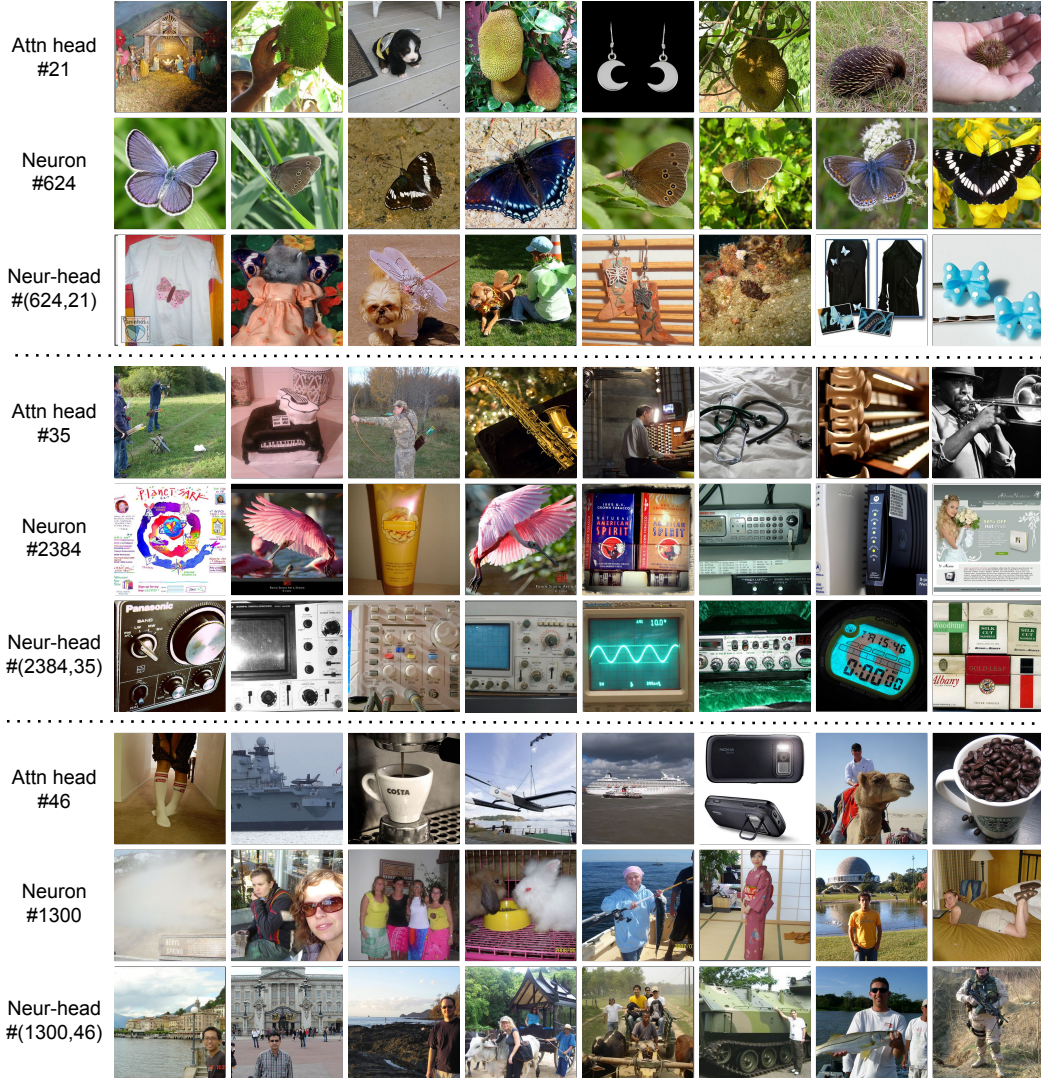


Figure 3: **Images with largest contribution norm for attention heads, neurons, and neuron-head pairs.** We present the top images from ImageNet validation set. Neuron-head pairs correspond to specific subcategory concepts of neurons (e.g., ‘butterfly *clothing*’ in row 3) and similar concepts to their neurons (e.g., ‘router’ for neuron #2384 and ‘people’ for neuron #1300)

193 using $m = 64$ text descriptions for neuron-head pairs surpasses the neuron baseline in reconstruction
 194 accuracy, exemplifying the benefits of our fine-grained decomposition.

195 **Comparison to neuron contribution.** We repeat the experiments above for r^n . To include the
 196 variance explained by multiple principal components, we simply reconstruct from a set $\{\hat{r}_1^n, \hat{r}_2^n, \dots\}$,
 197 where $\hat{r}_k^n$ is the k -th principal component. We use $\hat{r}_1^n$ for all sparse decomposition experiments. We
 198 find that, in comparison to our neuron-attention finding above, the first principal component explains
 199 much less variance – it reconstructs with a $\sim 4\%$ drop in classification accuracy – and that the
 200 contributions are less sparse – our mean-ablation experiment shows a nearly 38% accuracy difference
 201 our mean-ablation experiment with the top 10% neurons and top 10% neuron-head pairs.

202 4.2 Qualitative analysis

203 **Image retrieval.** We find the images I across the ImageNet validation dataset that give the highest
 204 contribution $r^h(I)$, $r^n(I)$, or $r^{n,h}(I)$ in norm. In Figure 3, we select neuron-head pair #624, 21 as

Components	Text descriptions
Attention head #21	“slr”, “vantage”, “jetta”
Neuron #624	“wings” (+1.73), “butterfly” (+1.69), “kite” (+1.20)
Neuron-head #(624, 21)	“butterfly” (+2.64), “roses” (−1.17), “grizzlies” (+0.91)
Attention head #35	“musicians”, “motorcycles”, “archery”
Neuron #2384	“kobe” (+1.00), “alfa” (−0.98), “redmond” (+0.91)
Neuron-head #(2384,35)	“routers” (+2.10), “tutorials” (−0.99), “spur” (−0.94)
Attention head #46	“salford”, “jcpenney”, “chattanooga”
Neuron #1300	“smileys” (+1.60), “masterpieces” (+1.31), “affiliated” (+1.16)
Neuron-head #(1300,46)	“cnn” (+1.26), “cto” (−1.25), “varsity” (−0.99)

Table 3: **Sparse text-based decomposition examples for Figure 3 components (RN50x16).** We select from the descriptions detailed in Section 4.1 with sparsity $m = 64$ and use TextSpan (Gandelman et al. [2024b]) to obtain descriptions for each attention head.

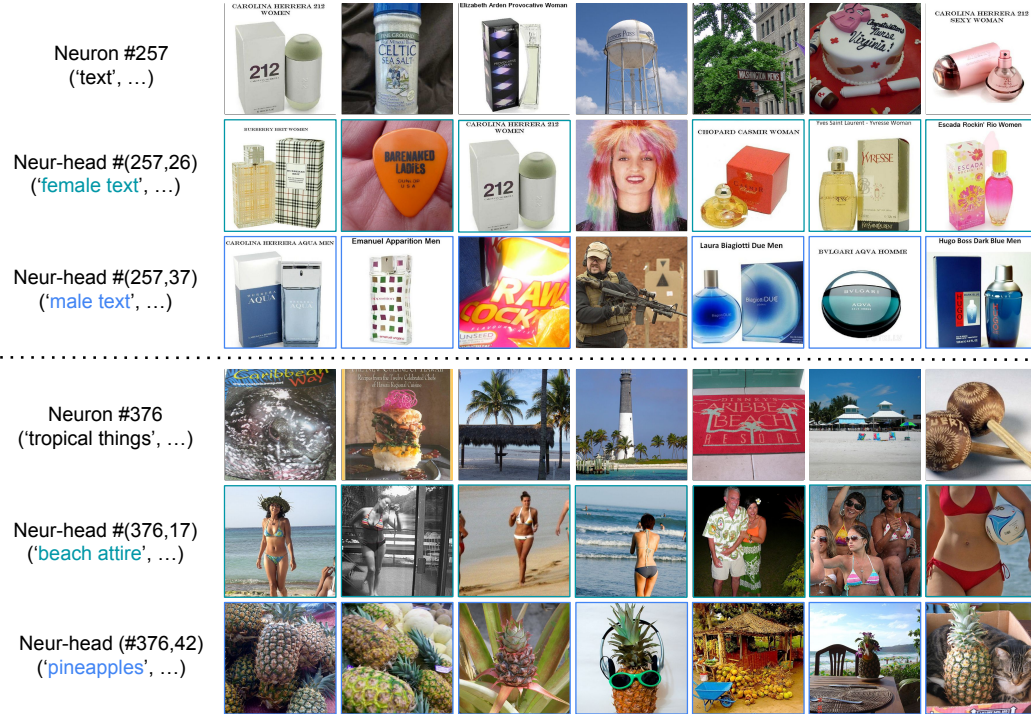


Figure 4: **Sub-concept neuron-head pairs for select neurons.** We again present top images by contribution norm and depict sub-concept relationships for two different neurons.

the pair with the top cosine similarity to the embedding of the word “butterfly”, and choose the other two pairs from four randomly selected pairs. We note that, as shown in the figure, the top-activating images for a given neuron-head pair look similar to those of its corresponding neuron, but not its corresponding attention head. Notably, the neuron #624 is most active on images that represent the ‘butterfly’ concept, and the neuron-head pair #(624,21) is most active for the *sub-concept* ‘butterfly clothing’. These examples highlight the ability of neuron-attention decomposition to isolate semantically meaningful directions in CLIP’s embedding space.

Neuron-head pairs remain polysemantic. We

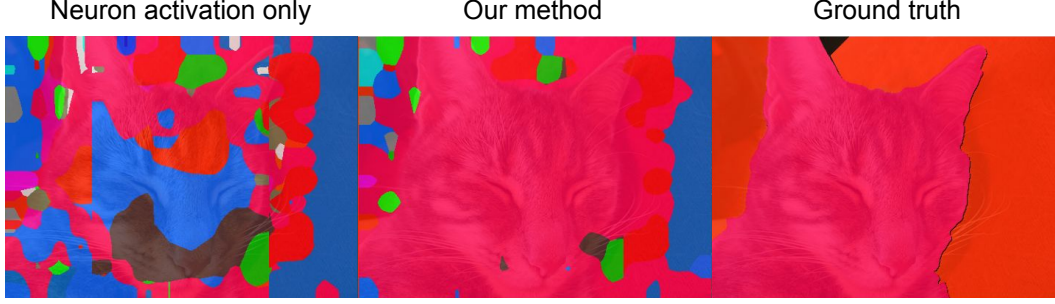


Figure 5: **Semantic segmentation using neuron-head pairs.** Multiplying the neuron-based heatmaps by the head-based heatmaps reduces the noise present in the segmentation map of a single component.

213 **Finding sub-concepts.**

214 5 Applications

215 5.1 Semantic segmentation

216 Existing literature on using interpretability for segmentation (Gandelsman et al. [2024b], Gandelsman
 217 et al. [2024a], Helbling et al. [2025]) mainly focuses on ImageNet-Segmentation (Guillaumin et al.
 218 [2014]) or other single-class tasks. We aim to show the viability of interpretability for dense segmen-
 219 tation tasks, validating our approach in the paradigm of training-free CLIP semantic segmentation
 220 (Appendix A.2). Unlike related works in this area, we do not alter any computation within CLIP (i.e.,
 221 we do not use self-self attention), and we use CLIP’s actual output for text-based cosine similarity
 222 (i.e., we decompose and use the output class token rather than the other K processed tokens).

223 **Method.** Our method begins by collecting two separate features from the model: the final-layer
 224 activation map $Z(I)$, which we denote as $L_{\text{act}}(I) \in \mathbb{R}^{H' \times W' \times C}$, and the per-head contributions from
 225 each of the $K = H'W'$ image patches $\{r_i^h(I)\}_{i \in \{1, \dots, K\}}$. We compute the per-head segmentation
 226 heatmap $L_{\text{sim}}(I)$ for class name t_j by calculating $\langle r_i^h(I), M_{\text{text}}(t_j) \rangle$ individually for all tokens i and
 227 heads h , and then aggregating these similarity maps such that $L_{\text{sim}}(I) \in \mathbb{R}^{H' \times W' \times H}$.

228 Next, we refine $L_{\text{act}}(I)$ and $L_{\text{sim}}(I)$ using our neuron-attention decomposition. Selecting the top-
 229 k neuron-head pairs by cosine similarity to class t_j and denoting the ordering of these pairs by
 230 $(n_1, h_1), (n_2, h_2), \dots, (n_k, h_k)$, we finally compute the segmentation logits for class t_j as:

$$\hat{L}(I) = \sum_{r=1}^k L_{\text{act}}^{n_r}(I) \circ L_{\text{sim}}^{h_r}(I) \quad (6)$$

231 where the superscripts n_r and h_r denote the top r -th neuron thead.

232 **Implementation details.** We evaluate our method on the PASCAL Context dataset (Mottaghi et al.
 233 [2014]). Similar to related works, we adopt a slide inference image pre-processing approach, where
 234 we specifically resize images to have a shorter side of 512 and then use a 384×384 window with
 235 a 192 stride. Additionally, we do not modify the class names in any way, both to select our top- k
 236 neuron-head pairs and to compute cosine similarity.

237 We report all results by selecting the top $k = 20000$ neuron-head pairs to text. We defer the effects of
 238 varying k , as well as the effect of intervening on register neurons (Darcet et al. [2023], Jiang et al.
 239 [2025]), to Appendix A.3.

240 **Main results.** As shown in Table 4, our method outperforms previous methods for semantic
 241 segmentation using CLIP-ResNet. To ensure fair comparison, we evaluate MaskCLIP on the same
 242 slide inference setup detailed in the previous paragraph, and use the ResNet50x16 backbone across
 243 all our experiments. We also present the results reported by SC-CLIP (Bai et al. [2024]), which is
 244 the current state-of-the-art method that uses CLIP-ViT – however, we stress that SC-CLIP leverages
 245 self-self attention, which, as shown, performs poorly when applied to CLIP-ResNet.

Method	mIoU(%)	Backbone
Self-self	22.2	RN50x16
MaskCLIP	22.8	RN50x16
Ours	26.2	RN50x16
SC-CLIP	40.1	ViT-B/16

Table 4: **Semantic segmentation performance on PASCAL Context (RN50x16).** For fair comparison, we implement MaskCLIP and $QQ^T + KK^T$ attention on the same slide inference setup we use. We gray out SC-CLIP, the current state-of-the-art, due to its ViT backbone.

Method	mIoU (%)
Neuron maps only	16.5
Head maps only	24.7
Both (multiplied)	26.2

Table 5: **Comparing different decomposition-based methods.** Segmentation by multiplying neuron features by head features performs better than by solely heads or solely neurons.

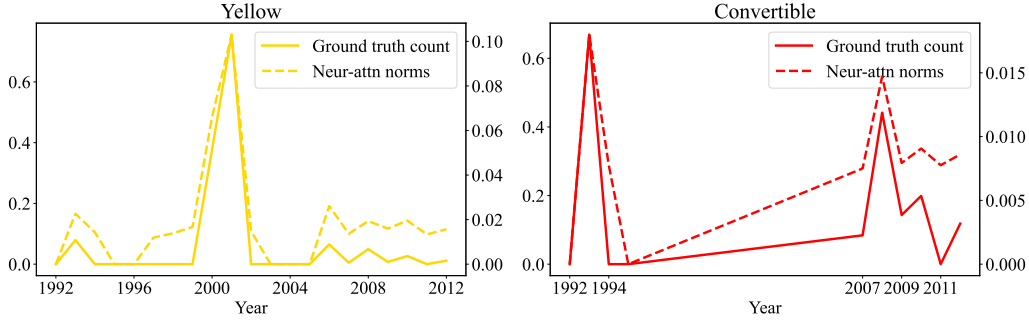


Figure 6: **Monitoring the distribution shift of Stanford Cars.** We compare the ground truth concept prevalence (left y-axis) to the proportional contribution (in norm) of top neuron-head pairs for a given concept (right y-axis).

We additionally show in Table 5 that our segmentation method is better than performing segmentation with a single aggregated segmentation map, verifying the efficacy of our neuron-attention decomposition over previous decomposition methods.

5.2 Monitoring distribution shift

Following Bhalla et al. [2024], we consider the Stanford Cars dataset (Krause et al. [2013]) and monitor the distribution shift of various car categories over time, shown in Figure 6. Specifically, we choose the top $k = 5$ neuron-head pairs for a concept by cosine similarity, and then compute these pairs' contribution norm divided by the contribution norm of the model output itself. We compare this to the ground truth proportion of concepts per-year. All of these concept contributions, along with tens of thousands more we don't analyze, are collected in a single forward pass per-image. This fact combined with our depicted results indicate that, with better understanding and selection of neuron-head pairs, our method may be scalable to real-world settings.

6 Limitations, discussion, and future work

We conclude by presenting two limitations of our approach and discussing future work.

Analyzing previous layers. Earlier convolutional blocks could give us a more complete understanding of the model's computation – the neurons of the last layer give insights into the final classification, but limited information on how that prediction was iteratively formed. This is especially relevant for semantic segmentation, where related CLIP-ViT methods leverage the spatial consistency of intermediate layers for improved performance.

Neuron-head pairs remain polysemantic.

Linear representation hypothesis?

267 **Future work.**

268 **References**

- 269 Sule Bai, Yong Liu, Yifei Han, Haoji Zhang, and Yansong Tang. Self-calibrated clip for training-free
270 open-vocabulary segmentation. *arXiv preprint arXiv:2411.15869*, 2024.
- 271 David Bau, Bolei Zhou, Aditya Khosla, Aude Oliva, and Antonio Torralba. Network dissection:
272 Quantifying interpretability of deep visual representations. In *Proceedings of the IEEE Conference*
273 *on Computer Vision and Pattern Recognition (CVPR)*, July 2017.
- 274 Usha Bhalla, Alex Oesterling, Suraj Srinivas, Flavio P. Calmon, and Himabindu Lakkaraju. In-
275 terpreting clip with sparse linear concept embeddings (splice). In A. Globerson, L. Mackey,
276 D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang, editors, *Advances in Neu-*
277 *ral Information Processing Systems*, volume 37, pages 84298–84328. Curran Associates,
278 Inc., 2024. URL [https://proceedings.neurips.cc/paper_files/paper/2024/file/](https://proceedings.neurips.cc/paper_files/paper/2024/file/996bef37d8a638f37bdfcac2789e835d-Paper-Conference.pdf)
279 [996bef37d8a638f37bdfcac2789e835d-Paper-Conference.pdf](https://proceedings.neurips.cc/paper_files/paper/2024/file/996bef37d8a638f37bdfcac2789e835d-Paper-Conference.pdf).
- 280 Steven Bills, Nick Cammarata, Dan Mossing, Henk Tillman, Leo Gao, Gabriel Goh, Ilya Sutskever,
281 Jan Leike, Jeff Wu, and William Saunders. Language models can explain neurons in language
282 models. [https://openaipublic.blob.core.windows.net/neuron-explainer/paper/](https://openaipublic.blob.core.windows.net/neuron-explainer/paper/index.html)
283 [index.html](https://openaipublic.blob.core.windows.net/neuron-explainer/paper/index.html), 2023.
- 284 Walid Boussethem, Felix Petersen, Vittorio Ferrari, and Hilde Kuehne. Grounding everything: Emerg-
285 ing localization properties in vision-language transformers. *arXiv preprint arXiv:2312.00878*,
286 2023.
- 287 Nick Cammarata, Shan Carter, Gabriel Goh, Chris Olah, Michael Petrov, Ludwig Schubert, Chelsea
288 Voss, Ben Egan, and Swee Kiat Lim. Thread: Circuits. *Distill*, 2020. doi: 10.23915/distill.00024.
289 <https://distill.pub/2020/circuits>.
- 290 Timothée Darcet, Maxime Oquab, Julien Mairal, and Piotr Bojanowski. Vision transformers need
291 registers. *arXiv preprint arXiv:2309.16588*, 2023.
- 292 Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale
293 hierarchical image database. In *Computer Vision and Pattern Recognition, 2009. CVPR 2009.*
294 *IEEE Conference on*, pages 248–255. IEEE, 2009. URL [https://ieeexplore.ieee.org/](https://ieeexplore.ieee.org/abstract/document/5206848/)
295 [abstract/document/5206848/](https://ieeexplore.ieee.org/abstract/document/5206848/).
- 296 Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda
297 Askell, Yuntao Bai, Anna Chen, Tom Conerly, Nova DasSarma, Dawn Drain, Deep Ganguli,
298 Zac Hatfield-Dodds, Danny Hernandez, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal
299 Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, and Chris
300 Olah. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 2021.
301 <https://transformer-circuits.pub/2021/framework/index.html>.
- 302 Yossi Gandelsman, Alexei A. Efros, and Jacob Steinhardt. Interpreting the second-order effects of
303 neurons in clip. *ArXiv*, abs/2406.04341, 2024a. URL [https://api.semanticscholar.org/](https://api.semanticscholar.org/CorpusID:270285780)
304 [CorpusID:270285780](https://api.semanticscholar.org/CorpusID:270285780).
- 305 Yossi Gandelsman, Alexei A. Efros, and Jacob Steinhardt. Interpreting clip’s image representation
306 via text-based decomposition, 2024b. URL <https://arxiv.org/abs/2310.05916>.
- 307 Matthieu Guillaumin, Daniel Küttel, and Vittorio Ferrari. Imagenet auto-annotation with segmentation
308 propagation. *International Journal of Computer Vision*, 110(3):328–348, 2014.
- 309 Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep Residual Learning for Im-
310 age Recognition. In *Proceedings of 2016 IEEE Conference on Computer Vision and Pattern*
311 *Recognition, CVPR ’16*, pages 770–778. IEEE, June 2016. doi: 10.1109/CVPR.2016.90. URL
312 <http://ieeexplore.ieee.org/document/7780459>.
- 313 Alec Helbling, Tuna Han Salih Meral, Ben Hoover, Pinar Yanardag, and Duen Horng Chau.
314 Conceptattention: Diffusion transformers learn highly interpretable features. *arXiv preprint*
315 *arXiv:2502.04320*, 2025.

316 Evan Hernandez, Sarah Schwettmann, David Bau, Teona Bagashvili, Antonio Torralba, and Jacob
317 Andreas. Natural language descriptions of deep visual features. *CoRR*, abs/2201.11114, 2022.
318 URL <https://arxiv.org/abs/2201.11114>.

319 Chao Jia, Yinfei Yang, Ye Xia, Yi-Ting Chen, Zarana Parekh, Hieu Pham, Quoc V. Le, Yun-Hsuan
320 Sung, Zhen Li, and Tom Duerig. Scaling up visual and vision-language representation learning
321 with noisy text supervision. In *International Conference on Machine Learning*, 2021. URL
322 <https://api.semanticscholar.org/CorpusID:231879586>.

323 Nick Jiang, Amil Dravid, Alexei Efros, and Yossi Gandelsman. Vision transformers don’t need
324 trained registers, 2025. URL <https://arxiv.org/abs/2506.08010>.

325 Jonathan Krause, Michael Stark, Jia Deng, and Li Fei-Fei. 3d object representations for fine-grained
326 categorization. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*
327 *Workshops*, June 2013.

328 Victor Lecomte, Kushal Thaman, Rylan Schaeffer, Naomi Bashkansky, Trevor Chow, and Sanmi
329 Koyejo. What causes polysemanticity? an alternative origin story of mixed selectivity from
330 incidental causes, 2024. URL <https://arxiv.org/abs/2312.03096>.

331 Jack Lindsey, Wes Gurnee, Emmanuel Ameisen, Brian Chen, Adam Pearce, Nicholas L. Turner,
332 Craig Citro, David Abrahams, Shan Carter, Basil Hosmer, Jonathan Marcus, Michael Sklar, Adly
333 Templeton, Trenton Bricken, Callum McDougall, Hoagy Cunningham, Thomas Henighan, Adam
334 Jermyn, Andy Jones, Andrew Persic, Zhenyi Qi, T. Ben Thompson, Sam Zimmerman, Kelley
335 Rivoire, Thomas Conerly, Chris Olah, and Joshua Batson. On the biology of a large language
336 model. *Transformer Circuits Thread*, 2025. URL [https://transformer-circuits.pub/](https://transformer-circuits.pub/2025/attribution-graphs/biology.html)
337 [2025/attribution-graphs/biology.html](https://transformer-circuits.pub/2025/attribution-graphs/biology.html).

338 Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. In
339 A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, editors, *Advances in*
340 *Neural Information Processing Systems*, volume 36, pages 34892–34916. Curran Associates,
341 Inc., 2023. URL [https://proceedings.neurips.cc/paper_files/paper/2023/file/](https://proceedings.neurips.cc/paper_files/paper/2023/file/6dcf277ea32ce3288914faf369fe6de0-Paper-Conference.pdf)
342 [6dcf277ea32ce3288914faf369fe6de0-Paper-Conference.pdf](https://proceedings.neurips.cc/paper_files/paper/2023/file/6dcf277ea32ce3288914faf369fe6de0-Paper-Conference.pdf).

343 Roozbeh Mottaghi, Xianjie Chen, Xiaobai Liu, Nam-Gyu Cho, Seong-Whan Lee, Sanja Fidler, Raquel
344 Urtasun, and Alan Yuille. The role of context for object detection and semantic segmentation in
345 the wild. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2014.

346 Neel Nanda, Lawrence Chan, Tom Lieberum, Jess Smith, and Jacob Steinhardt. Progress measures
347 for grokking via mechanistic interpretability. *arXiv preprint arXiv:2301.05217*, 2023.

348 Tuomas Oikarinen and Tsui-Wei Weng. Linear explanations for individual neurons. *arXiv preprint*
349 *arXiv:2405.06855*, 2024.

350 Yagyensh C. Pati, Ramin Rezaifar, and Perinkulam S. Krishnaprasad. Orthogonal matching pursuit:
351 recursive function approximation with applications to wavelet decomposition. *Proceedings of*
352 *27th Asilomar Conference on Signals, Systems and Computers*, pages 40–44 vol.1, 1993. URL
353 <https://api.semanticscholar.org/CorpusID:16513805>.

354 Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal,
355 Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever.
356 Learning transferable visual models from natural language supervision. In *Proceedings of the 38th*
357 *International Conference on Machine Learning (ICML)*, volume 139, pages 8748–8763. PMLR,
358 2021.

359 Aditya Ramesh, Mikhail Pavlov, Gabriel Goh, Scott Gray, Chelsea Voss, Alec Radford, Mark Chen,
360 and Ilya Sutskever. Zero-shot text-to-image generation, 2021. URL [https://arxiv.org/abs/](https://arxiv.org/abs/2102.12092)
361 [2102.12092](https://arxiv.org/abs/2102.12092).

362 Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-
363 resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF confer-*
364 *ence on computer vision and pattern recognition*, pages 10684–10695, 2022.

- 365 Tong Shao, Zhuotao Tian, Hang Zhao, and Jingyong Su. Explore the potential of clip for training-free
 366 open vocabulary semantic segmentation. In *European Conference on Computer Vision*, pages
 367 139–156. Springer, 2024.
- 368 Lee Sharkey, Bilal Chughtai, Joshua Batson, Jack Lindsey, Jeff Wu, Lucius Bushnaq, Nicholas
 369 Goldowsky-Dill, Stefan Heimersheim, Alejandro Ortega, Joseph Bloom, Stella Biderman, Adria
 370 Garriga-Alonso, Arthur Conmy, Neel Nanda, Jessica Rumbelow, Martin Wattenberg, Nandi
 371 Schoots, Joseph Miller, Eric J. Michaud, Stephen Casper, Max Tegmark, William Saunders, David
 372 Bau, Eric Todd, Atticus Geiger, Mor Geva, Jesse Hoogland, Daniel Murfet, and Tom McGrath.
 373 Open problems in mechanistic interpretability, 2025. URL [https://arxiv.org/abs/2501.](https://arxiv.org/abs/2501.16496)
 374 [16496](https://arxiv.org/abs/2501.16496).
- 375 Feng Wang, Jieru Mei, and Alan Yuille. Sclip: Rethinking self-attention for dense vision-language
 376 inference. In *Computer Vision – ECCV 2024: 18th European Conference, Milan, Italy, September*
 377 *29–October 4, 2024, Proceedings, Part XXI*, page 315–332, Berlin, Heidelberg, 2024. Springer-
 378 Verlag. ISBN 978-3-031-72663-7. doi: 10.1007/978-3-031-72664-4_18. URL [https://doi.](https://doi.org/10.1007/978-3-031-72664-4_18)
 379 [org/10.1007/978-3-031-72664-4_18](https://doi.org/10.1007/978-3-031-72664-4_18).
- 380 Mitchell Wortsman. Reaching 80% zero-shot accuracy with openclip: Vit-g/14 trained on laion-2b.
 381 <https://laion.ai/blog/giant-openclip/>, 2023. Accessed: 2025-08-12.
- 382 Xiaohua Zhai, Basil Mustafa, Alexander Kolesnikov, and Lucas Beyer. Sigmoid loss for language
 383 image pre-training, 2023. URL <https://arxiv.org/abs/2303.15343>.
- 384 Jitian Zhao, Chenghui Li, Frederic Sala, and Karl Rohe. Quantifying structure in clip embeddings:
 385 A statistical framework for concept interpretation, 2025. URL [https://arxiv.org/abs/2506.](https://arxiv.org/abs/2506.13831)
 386 [13831](https://arxiv.org/abs/2506.13831).
- 387 Chong Zhou, Chen Change Loy, and Bo Dai. Extract free dense labels from clip. In *European*
 388 *conference on computer vision*, pages 696–712. Springer, 2022.
- 389 Xiangyang Zhu, Renrui Zhang, Bowei He, Ziyu Guo, Ziyao Zeng, Zipeng Qin, Shanghang Zhang,
 390 and Peng Gao. Pointclip v2: Prompting clip and gpt for powerful 3d open-world learning. In *2023*
 391 *IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 2639–2650, 2023. doi:
 392 10.1109/ICCV51070.2023.00249.

393 A Appendix

394 A.1 Reconstruction details

395 Specifically, we take $\hat{r}^{n,h}$ to be the first principal component computed over the set $\mathcal{S}^{n,h}$, the top 50
 396 $r^{n,h}(I)$ samples from $\mathcal{D}$ by norm. We compute $\{\hat{r}_1^n, \hat{r}_2^n, \dots\}$ analogously.
 397 In Figure 2 and Table 3, rather than decomposing the $\hat{r}_1^n$ we obtain from $\mathcal{D}$, we decompose the first
 398 principle component obtained from the *top 100* neuron contribution samples by norm, selected from
 399 5000 images from ImageNet test.

400 A.2 Related semantic segmentation using CLIP

401 Many recent works use the inherent semantic alignment of CLIP’s image and text encoders for
 402 dense segmentation tasks, in which CLIP features are either processed as pseudo-labels to train
 403 another network or are used themselves for segmentation predictions (Zhou et al. [2022]). We
 404 focus on the latter training-free paradigm, where state-of-the-art methods incorporate a *self-self*
 405 *attention* mechanism (Bousselham et al. [2023], Wang et al. [2024], Shao et al. [2024], Bai et al.
 406 [2024]). In contrast to self-self attention methods that discard the class token and rewrite attention,
 407 our proposed segmentation method uses the (decomposed) class token and does not rearrange any
 408 internal mechanisms.

409 In comparison to CLIP-ViT, CLIP-ResNet has disadvantages for segmentation purposes: it has
 410 more aggressive spatial downsampling (a factor of 32 compared to 16) and, as shown earlier, is not
 411 conducive to self-self attention. However, CLIP-ResNet is uniquely able to process arbitrarily-sized
 412 images without resizing, can be better at processing larger images, and its attention pooling structure
 413 makes our findings relevant to CLIP-ViT.

414 A.3 Semantic segmentation using neuron-attention decomposition

415 **Effect of varying k .** We observe that cosine similarity to text decreases exponentially as k increases –
416 intuitively, this should

417 **Effect of register neurons.**

418 Modifying select *register neurons* at test time was recently proposed (Jiang et al. [2025]) as an
419 alternative to trained registers (Darcet et al. [2023]) that mitigate irregular attention patterns. We
420 find a sparse set of register neurons for CLIP-ResNet and adopt the intervention detailed by Jiang
421 et al. [2025] – this intervention improves mIoU by 0.36% for our method using $k = 100$, but the
422 advantage becomes negligible or even detrimental with higher k . We note that our method of selecting
423 CLIP-ResNet’s register neurons hinges on the empirical observation, discussed in the next paragraph,
424 that CLIP-ResNet’s attention sink invariably appears in the last image token – a better register neuron
425 selection method may show further improvements.

426 **Finding register neurons.**

427 **CLIP-ResNet attention sink.**

428 Inspired by the “butterfly clothing” neuron-head pair $\#(624, 21)$ in Figure 3, we search for more
429 examples of sub-concept relationships. We find that our sparse text descriptions do not reliably
430 indicate sub-concepts (e.g., we do not find a ‘clothing’-like description in the sparse decomposition
431 of $\hat{r}^{624,21}$). Additionally, while

432 To narrow the search space, we only consider the 30 neurons whose cosine similarity $\langle \hat{r}^n, M_{\text{text}}(t_j) \rangle$
433 is above a certain threshold for any j , where $j \in \{1, 30000\}$ indexes the 30k most common English
434 words. We also

435 We stress that, although we only present findings for two neurons in Figure 4, we find several sub-
436 concept pairs for all six randomly selected neurons we inspect. Altogether, this shows the fine-grained
437 nature of neuron-attention decomposition, and